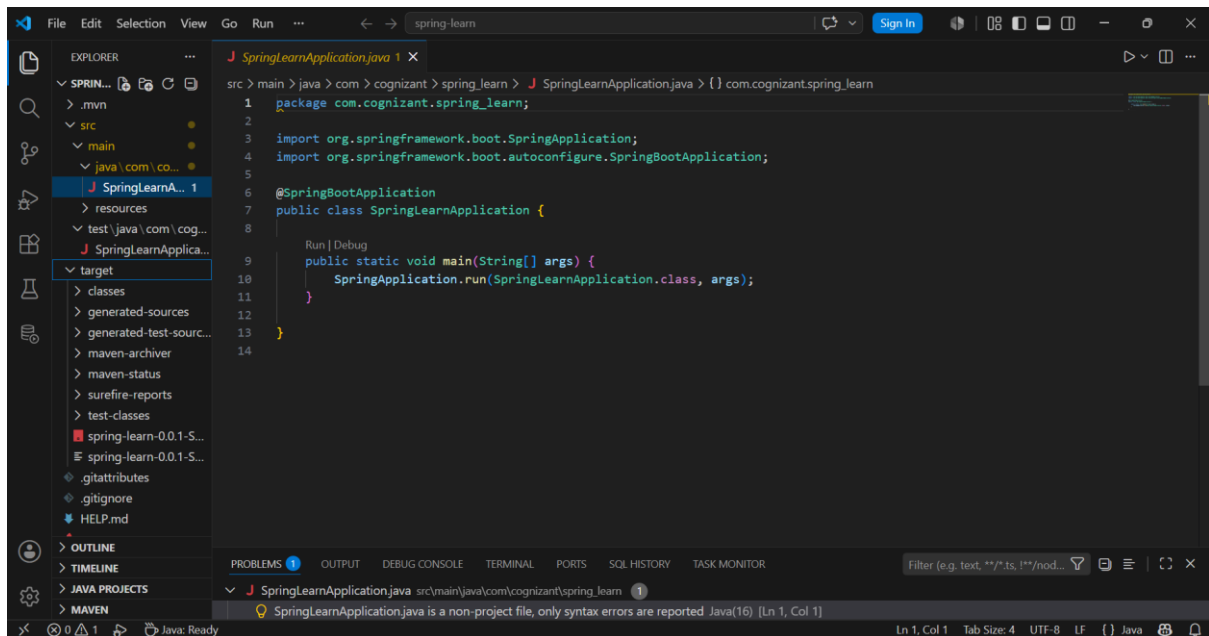
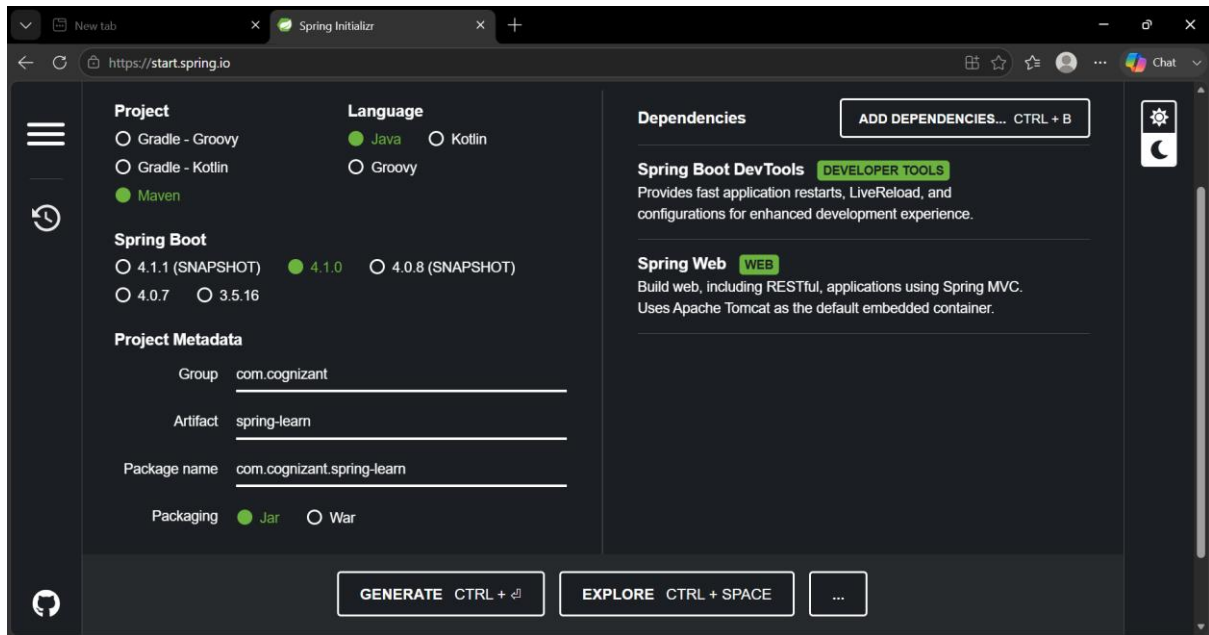


DIGITAL NURTURE 5.0 DEEPSKILLING

WEEK 3 – HANDS-ON

Spring REST using Spring Boot 3

Create a Spring Web Project using Maven



This screenshot shows an IDE window with the 'spring-learn' project open. The Explorer pane on the left shows the project structure, including 'src/main/resources' and 'test/java/com...'. The Terminal pane on the right shows the output of the 'mvn spring-boot:run' command. The output includes messages from Maven, Spring Boot, and the application's main method. The application is running on port 8080.

```
sri@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:spring-learn >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
>>> spring-boot:4.1.0:run (default-cli) > test-compile @ spring-learn >>>
[INFO]
--- resources:3.5.0:resources (default-resources) @ spring-learn ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
--- compiler:3.15.0:compile (default-compile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
--- resources:3.5.0:testResources (default-testResources) @ spring-learn ---
[INFO] skip non existing resourceDirectory C:\Users\sri\Downloads\spring-learn\spring-learn\src\test\resources
[INFO]
--- compiler:3.15.0:testCompile (default-testCompile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
<<< spring-boot:4.1.0:run (default-cli) < test-compile @ spring-learn <<<
[INFO]
--- spring-boot:4.1.0:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []
```

This screenshot shows the same IDE window after the application has started. The Terminal pane now displays the application's output, including a large ASCII art message that reads 'WELCOME TO SPRING BOOT'. Below the message, the application logs show the startup process, including the initialization of the Spring Boot application and the Tomcat web server. The application is running on port 8080.

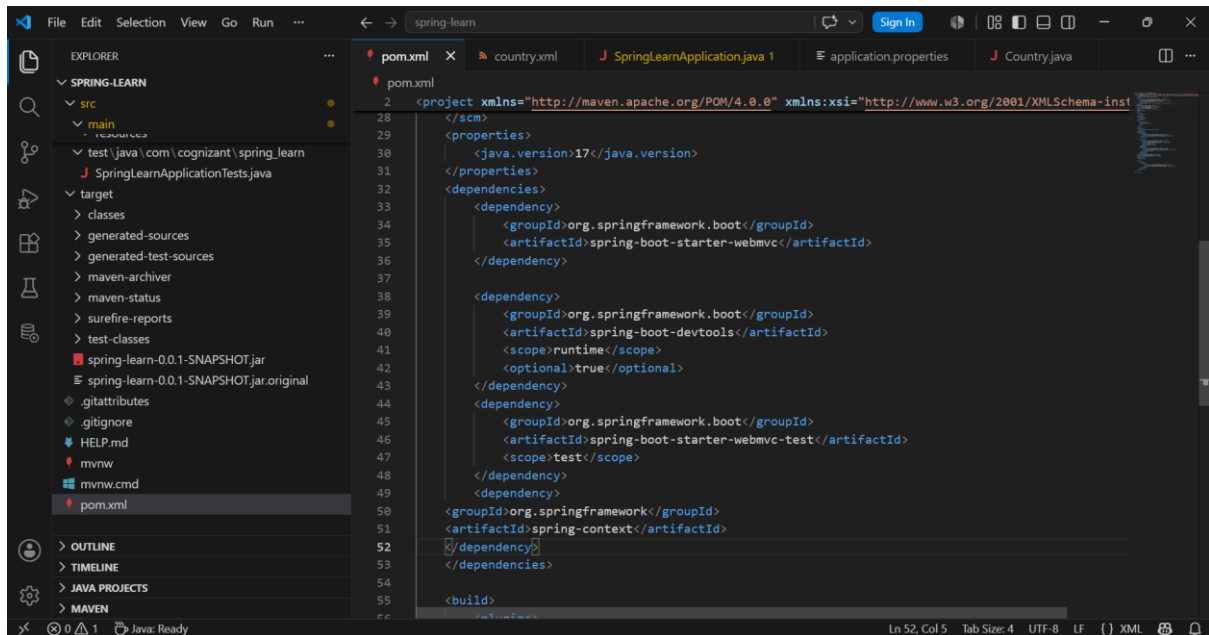
```
sri@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run

      .__  ___.
     / ____ \| | | |
    / ___/ /| | | |
   /  __/| |_| | | |
  /____/____|_|_|_|

:: Spring Boot ::      (v4.1.0)

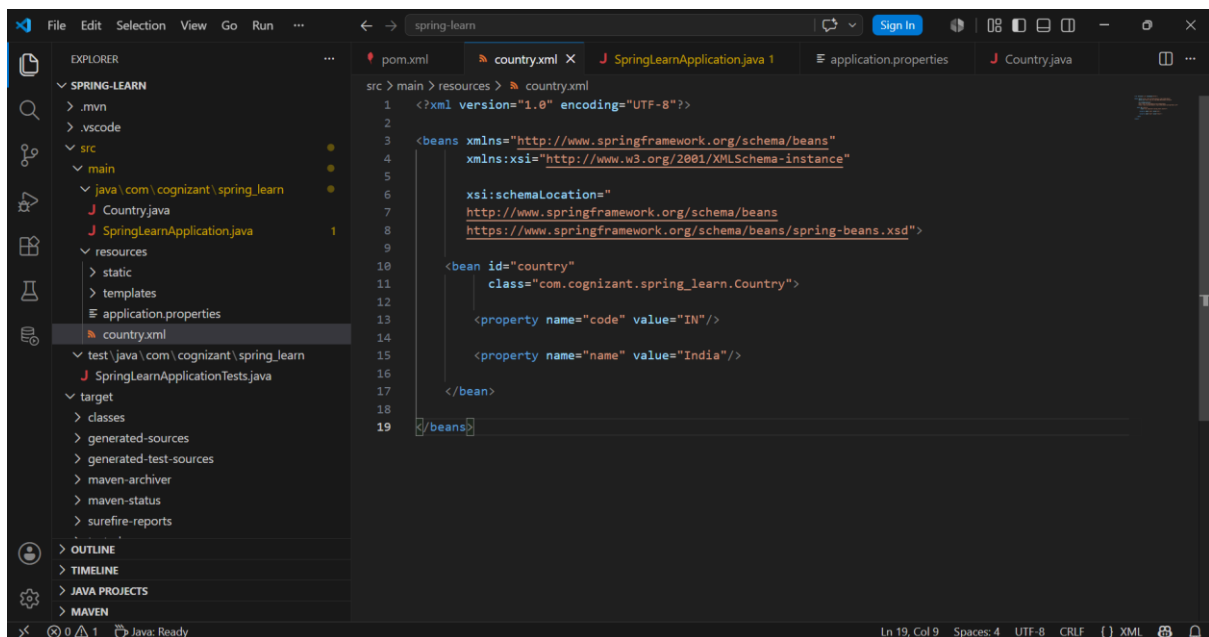
2026-06-29T21:56:14.622+05:30 INFO 3232 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting Spr
ingLearnApplication using Java 21.0.2 with PID 3232 (C:\Users\sri\Downloads\spring-learn\spring-learn\target\classes started by sri
ni in C:\Users\sri\Downloads\spring-learn\spring-learn)
2026-06-29T21:56:14.624+05:30 INFO 3232 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active pr
ofile set, falling back to 1 default profile: "default"
2026-06-29T21:56:14.706+05:30 INFO 3232 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools pro
perty defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-06-29T21:56:14.706+05:30 INFO 3232 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For addition
al web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-06-29T21:56:15.767+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initi
alized with port 8080 (http)
2026-06-29T21:56:15.784+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting ser
vice [Tomcat]
2026-06-29T21:56:15.784+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Ser
vlet engine: [Apache Tomcat/11.0.22]
2026-06-29T21:56:15.833+05:30 INFO 3232 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebAppl
icationContext: initialization completed in 1127 ms
2026-06-29T21:56:16.138+05:30 INFO 3232 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat start
ed on port 8080 (http) with context path '/'
2026-06-29T21:56:16.143+05:30 INFO 3232 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started Sri
ngLearnApplication in 2.21 seconds (process running for 2.578)
```

Spring Core – Load Country from Spring Configuration XML



This screenshot shows the Visual Studio Code editor with the `spring-learn` project open. The `EXPLORER` sidebar on the left displays the project structure, including `src/main/resources`. The main editor window shows the `pom.xml` file, which is a Maven Project Object Model (POM) file. The file contains dependencies for Spring Boot, Spring Boot DevTools, and Spring Context. The status bar at the bottom indicates the file is at line 52, column 5, with a tab size of 4, UTF-8 encoding, and LF line endings.

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.cognizant</groupId>
    <artifactId>spring-learn</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <packaging>jar</packaging>
    <name>Spring Learn</name>
    <description>Spring Learn</description>
    <properties>
        <java.version>17</java.version>
    </properties>
    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-webmvc</artifactId>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-devtools</artifactId>
            <scope>runtime</scope>
            <optional>true</optional>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-webmvc-test</artifactId>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-context</artifactId>
        </dependency>
    </dependencies>
    <build>
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
        </plugins>
    </build>
</project>
```

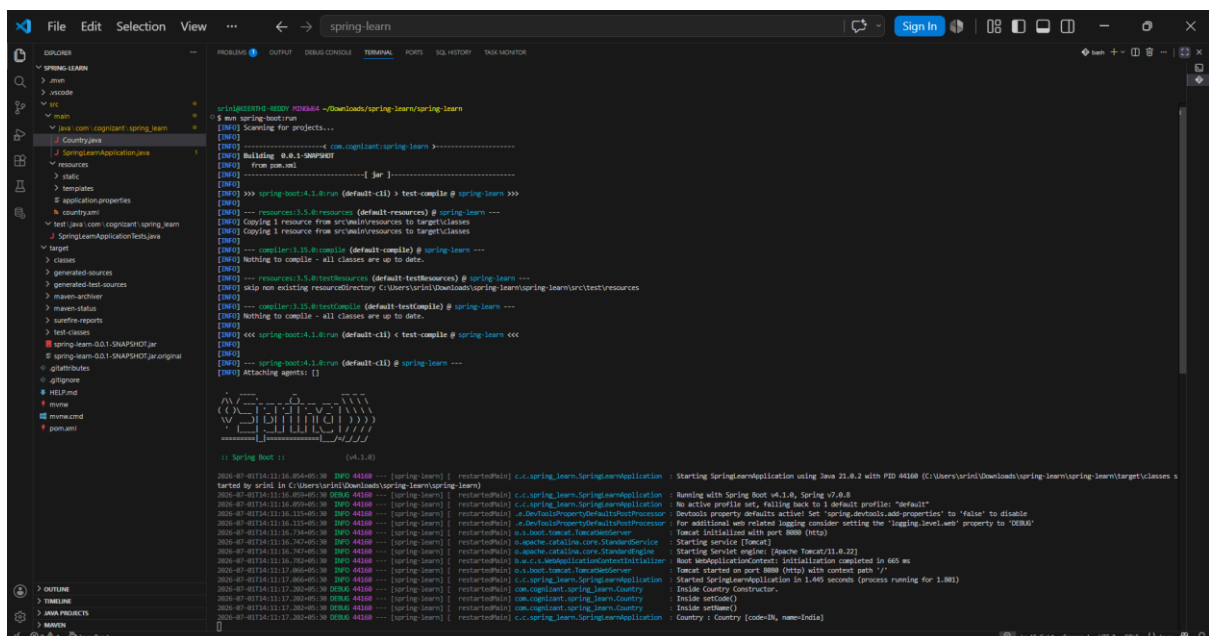
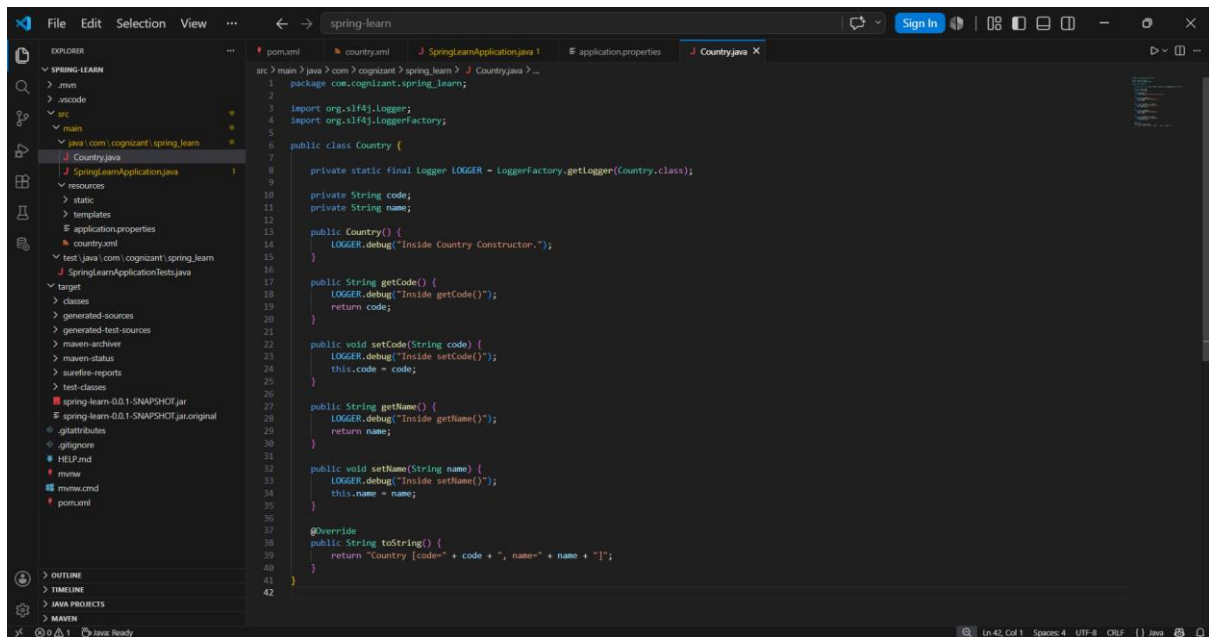


This screenshot shows the Visual Studio Code editor with the `spring-learn` project open. The `EXPLORER` sidebar on the left displays the project structure, including `src/main/resources`. The main editor window shows the `country.xml` file, which is a Spring XML configuration file. The file contains a bean definition for `Country` with properties `code` and `name`. The status bar at the bottom indicates the file is at line 19, column 9, with a tab size of 4, UTF-8 encoding, and CRLF line endings.

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://www.springframework.org/schema/beans
        http://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="country"
        class="com.cognizant.spring_learn.Country">
        <property name="code" value="IN"/>
        <property name="name" value="India"/>
    </bean>
</beans>
```

```
src > main > java > com > cognizant > spring_learn > J SpringLearnApplication.java > SpringLearnApplication
1 package com.cognizant.spring_learn;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8
9 import org.springframework.boot.SpringApplication;
10 import org.springframework.boot.autoconfigure.SpringBootApplication;
11
12 @SpringBootApplication
13 public class SpringLearnApplication {
14
15     // Logger declaration
16     private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class);
17
18     Run | Debug
19     public static void main(String[] args) {
20         SpringApplication.run(SpringLearnApplication.class, args);
21
22         // Call the method to load country bean
23         displayCountry();
24     }
25
26     public static void displayCountry() {
27
28         // Load the Spring XML configuration
29         ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
30
31         // Get the country bean
32         Country country = context.getBean("country", Country.class);
33
34         // Display country details
35         LOGGER.debug("Country : {}", country.toString());
36     }
37
38 }
```

```
src > main > resources > application.properties
1 spring.application.name=spring_learn
2 logging.level.com.cognizant.spring_learn=DEBUG
```



Hello World RESTful Web Service

This screenshot shows the Visual Studio Code editor with the `spring-learn` project open. The Explorer sidebar on the left displays the project structure, including the `src/main/java/com/cognizant/spring_learn/controller` directory. The `HelloController.java` file is selected and its content is displayed in the main editor. The code defines a `RestController` with a `sayHello()` method that logs and returns a message.

```
src > main > java > com > cognizant > spring_learn > controller > HelloController.java > ...
1 package com.cognizant.spring_learn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 import org.springframework.web.bind.annotation.GetMapping;
7 import org.springframework.web.bind.annotation.RestController;
8
9 @RestController
10 public class HelloController {
11
12     private static final Logger LOGGER =
13         LoggerFactory.getLogger(HelloController.class);
14
15     @GetMapping("/hello")
16     public String sayHello() {
17
18         LOGGER.info("START - sayHello()");
19         String message = "Hello World!!";
20
21         LOGGER.info("END - sayHello()");
22
23         return message;
24     }
25 }
26
27
28
```

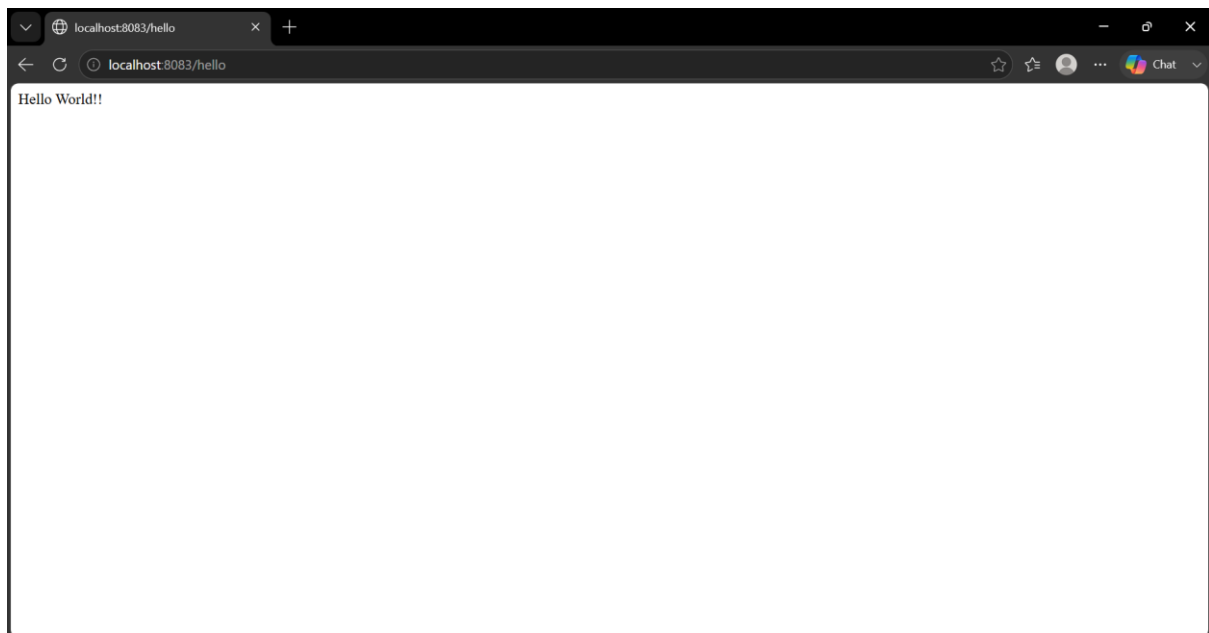
This screenshot shows the Visual Studio Code editor with the `spring-learn` project open. The Explorer sidebar on the left shows the `resources` directory selected. The `application.properties` file is open in the main editor, containing configuration for the application name, logging level, and server port.

```
src > main > resources > application.properties
1 spring.application.name=spring-learn
2 logging.level.com.cognizant.spring_learn=DEBUG
3 server.port=8083
4
```

```
src > main > resources > application.properties
1  spring.application.name=spring-learn
2  logging.level.com.cognizant.spring_learn=DEBUG
3  server.port=8083

srimi@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn clean package
OT-INF/.
[INFO] The original artifact has been renamed to C:\Users\srimi\Downloads\spring-learn\sp
ring-learn\target\spring-learn-0.0.1-SNAPSHOT.jar.original
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.472 s
[INFO] Finished at: 2026-07-01T14:24:28+05:30
[INFO] -----

srimi@KEERTHI-REDDY MINGW64 ~/Downloads/spring-learn/spring-learn
$ mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.cognizant:spring-learn >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
```



REST - Country Web Service

The screenshot shows the VS Code editor with the 'CountryController.java' file open. The file contains the following code:

```
src > main > java > com > cognizant > spring_learn > controller > CountryController.java > CountryController
1 package com.cognizant.spring_learn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8 import org.springframework.web.bind.annotation.RequestMapping;
9 import org.springframework.web.bind.annotation.RestController;
10
11 import com.cognizant.spring_learn.country;
12
13 @RestController
14 public class CountryController {
15
16     private static final Logger LOGGER =
17         LoggerFactory.getLogger(CountryController.class);
18
19     @RequestMapping("/country")
20     public Country getCountryIndia() {
21
22         LOGGER.info("START - getCountryIndia()");
23
24         ApplicationContext context =
25             new ClassPathXmlApplicationContext("country.xml");
26
27         Country country = context.getBean("country", Country.class);
28
29         LOGGER.info("END - getCountryIndia()");
30
31         return country;
32     }
33 }
```

The Explorer sidebar on the left shows the project structure, including 'pom.xml', 'country.xml', 'SpringLearnApplication.java', and 'CountryController.java'. The bottom status bar indicates 'Ln 33, Col 2'.

The screenshot shows the VS Code terminal with the output of a Spring Boot application. The output includes the following information:

```
spring@KEERTHI-REDDY MINGW64 ~/Downloads/spring_learn/spring_learn
$ mvn spring:boot:run
profile: 'default'
2026-07-01T14:32:06.211+05:30 INFO 23916 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.d
evtools.add-properties' to 'false' to disable
2026-07-01T14:32:06.211+05:30 INFO 23916 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider sett
ing the 'logging.level.web' property to 'DEBUG'
2026-07-01T14:32:06.863+05:30 INFO 23916 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-01T14:32:06.871+05:30 INFO 23916 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T14:32:06.871+05:30 INFO 23916 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-01T14:32:06.905+05:30 INFO 23916 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization compl
eted in 694 ms
2026-07-01T14:32:07.232+05:30 WARN 23916 --- [spring-learn] [ restartedMain] ConfigServletWebServerApplicationContext : Exception encountered during context initializ
ation - cancelling refresh attempt: org.springframework.context.ApplicationContextException: Failed to start bean 'webServerStartStop'
2026-07-01T14:32:07.242+05:30 INFO 23916 --- [spring-learn] [ restartedMain] .s.b.a.l.ConditionEvaluationReportLogger :

Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.
2026-07-01T14:32:07.267+05:30 ERROR 23916 --- [spring-learn] [ restartedMain] o.s.b.d.logging.FailureAnalysisReporter :

*****
APPLICATION FAILED TO START
*****

Description:

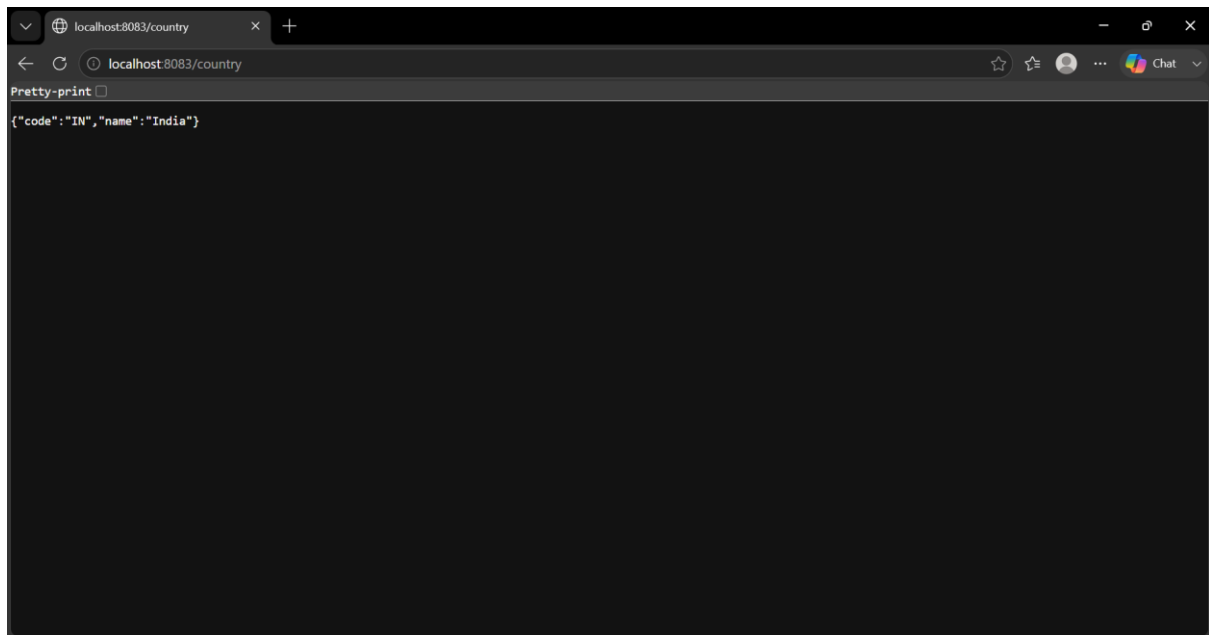
Web server failed to start. Port 8083 was already in use.

Action:

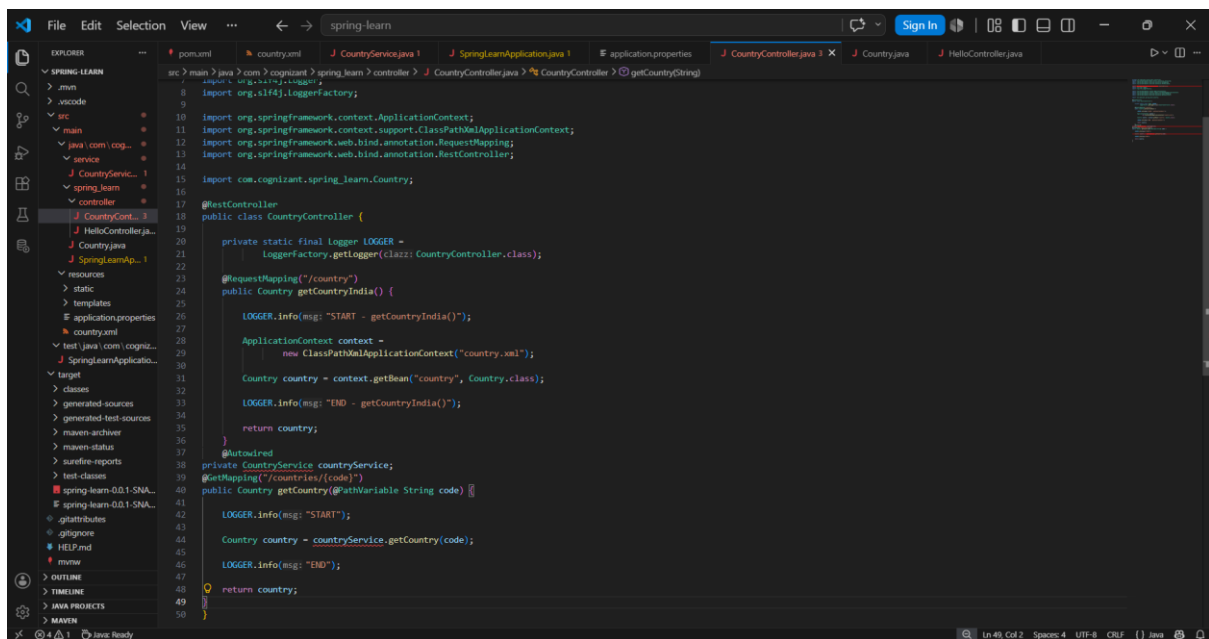
Identify and stop the process that's listening on port 8083 or configure this application to listen on another port.

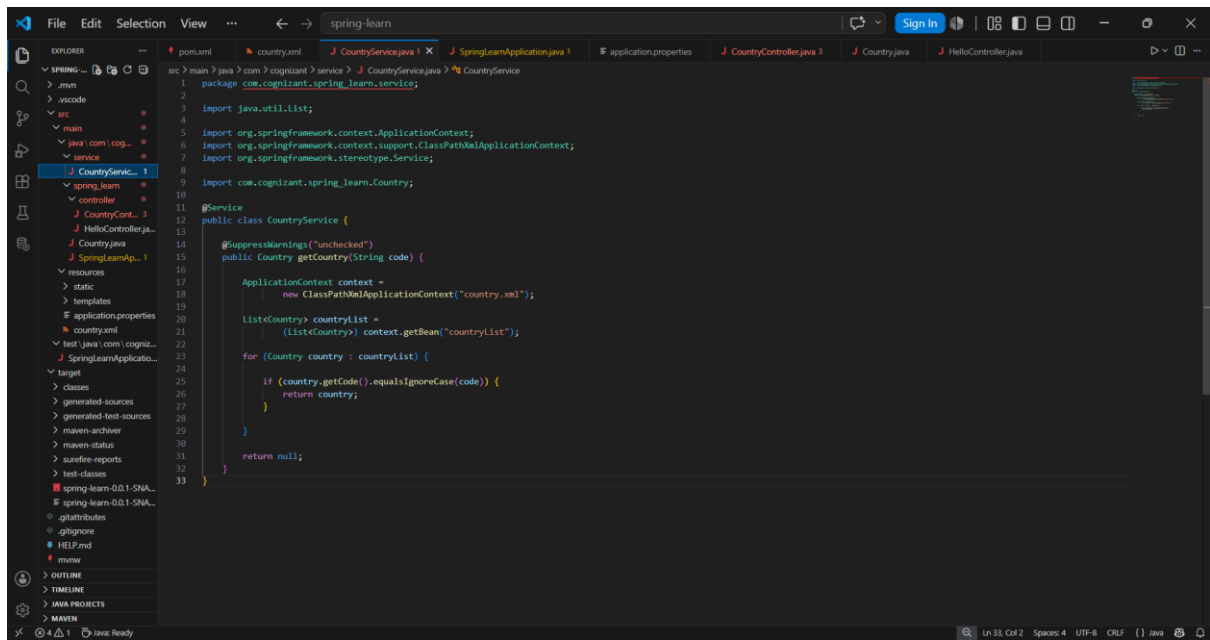
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.785 s
[INFO] Finished at: 2026-07-01T14:32:07+05:30
[INFO] -----
```

The terminal output shows the application starting successfully but failing due to a port conflict. The status bar at the bottom indicates 'Ln 20, Col 39'.

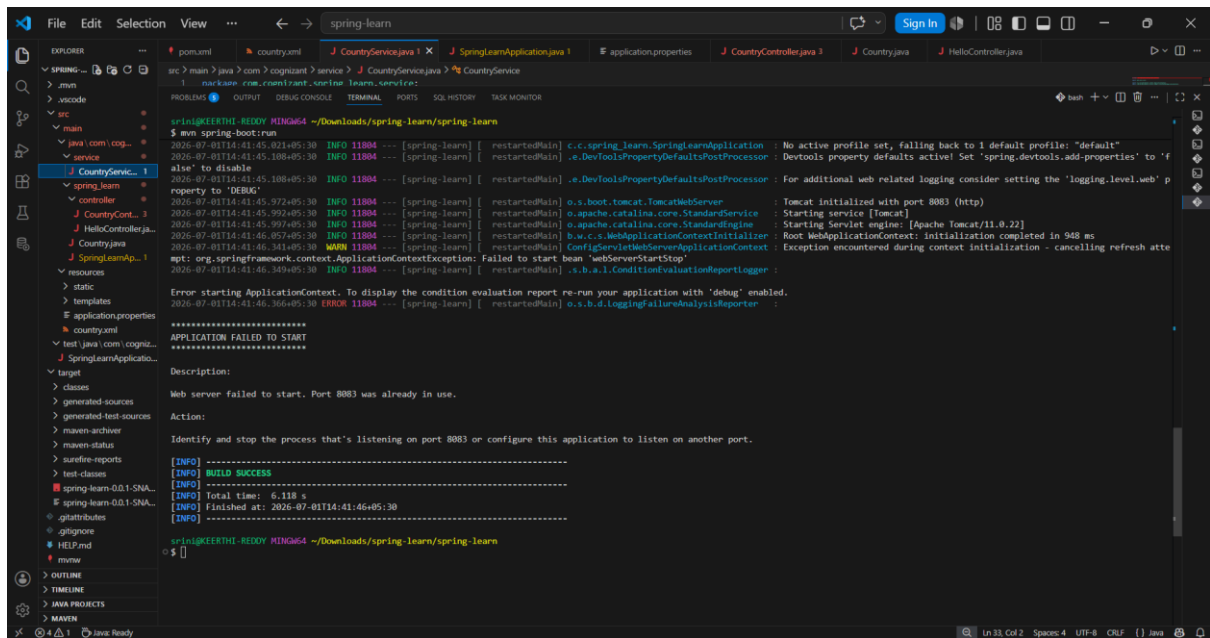


REST - Get country based on country code





```
src > main > java > com > cognizant > service > J CountryService.java > % CountryService
1 package com.cognizant.spring_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.stereotype.Service;
8
9 import com.cognizant.spring_learn.Country;
10
11 @Service
12 public class CountryService {
13
14     @SuppressWarnings("unchecked")
15     public Country getCountry(String code) {
16
17         ApplicationContext context =
18             new ClassPathXmlApplicationContext("country.xml");
19
20         List<Country> countryList =
21             (List<Country>) context.getBean("countryList");
22
23         for (Country country : countryList) {
24
25             if (country.getCode().equalsIgnoreCase(code)) {
26                 return country;
27             }
28
29         }
30
31         return null;
32     }
33 }
```



```
src > main > java > com > cognizant > service > J CountryService.java > % CountryService
1 package com.cognizant.spring_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.stereotype.Service;
8
9 import com.cognizant.spring_learn.Country;
10
11 @Service
12 public class CountryService {
13
14     @SuppressWarnings("unchecked")
15     public Country getCountry(String code) {
16
17         ApplicationContext context =
18             new ClassPathXmlApplicationContext("country.xml");
19
20         List<Country> countryList =
21             (List<Country>) context.getBean("countryList");
22
23         for (Country country : countryList) {
24
25             if (country.getCode().equalsIgnoreCase(code)) {
26                 return country;
27             }
28
29         }
30
31         return null;
32     }
33 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR

bash + - - - - -

```
spring@KEERTHI-REDDY HINGW64: ~/Downloads/spring_learn/spring_learn
$ mvn spring-boot:run
2026-07-01T14:41:45.021+05:30 INFO 11804 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-01T14:41:45.108+05:30 INFO 11804 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'f
alse' to disable
2026-07-01T14:41:45.108+05:30 INFO 11804 --- [spring-learn] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' p
roperty to 'DEBUG'
2026-07-01T14:41:45.272+05:30 INFO 11804 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-01T14:41:45.392+05:30 INFO 11804 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T14:41:45.397+05:30 INFO 11804 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-01T14:41:46.057+05:30 INFO 11804 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 948 ms
2026-07-01T14:41:46.341+05:30 WARN 11804 --- [spring-learn] [ restartedMain] ConfigServletWebServerApplicationContext : Exception encountered during context initialization - cancelling refresh atte
mpt: org.springframework.context.ApplicationContextException: Failed to start bean 'webServerStartStop'
2026-07-01T14:41:46.349+05:30 INFO 11804 --- [spring-learn] [ restartedMain] s.b.a.l.ConditionEvaluationReportLogger :

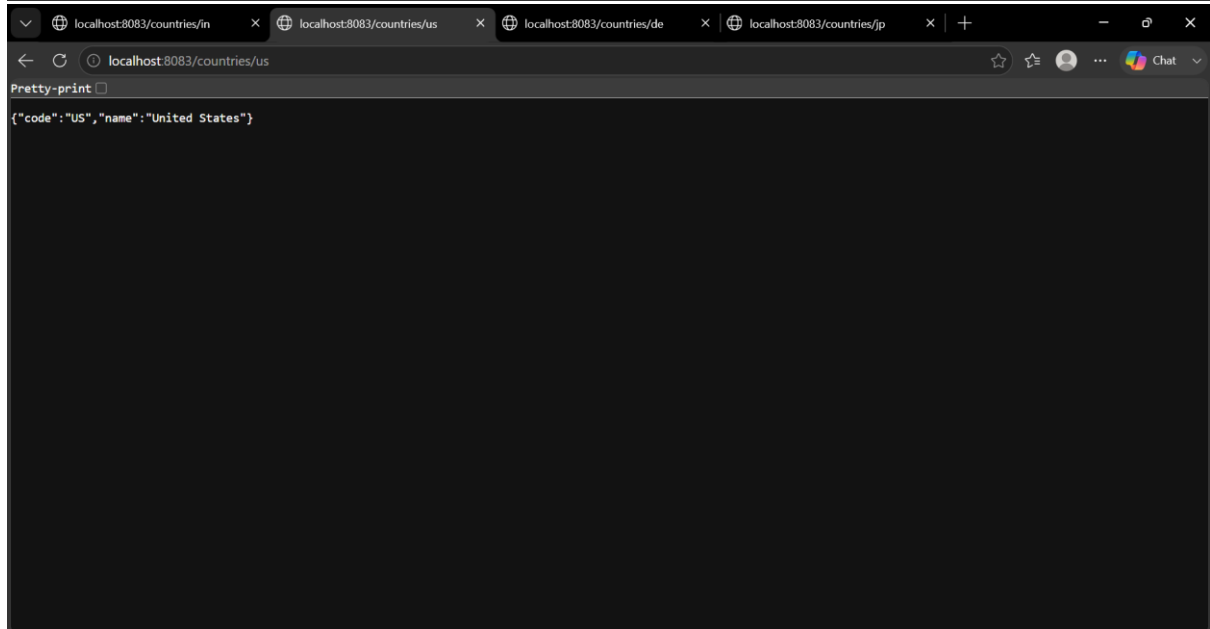
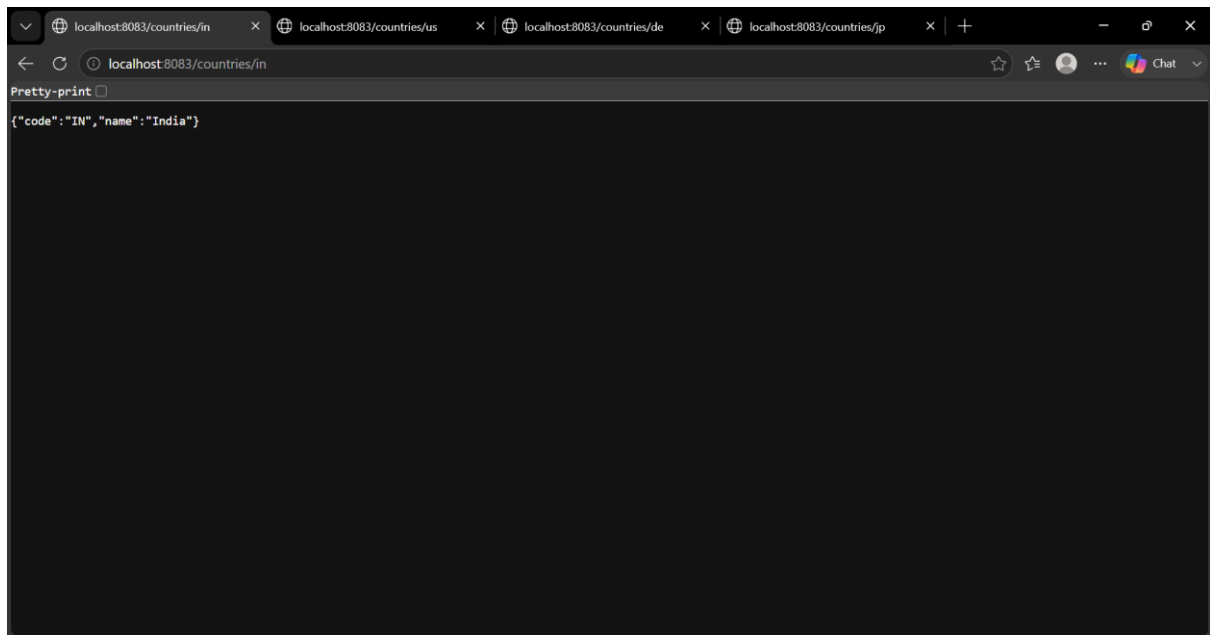
Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.
2026-07-01T14:41:46.366+05:30 ERROR 11804 --- [spring-learn] [ restartedMain] o.s.b.d.LoggingFailureAnalysisReporter :

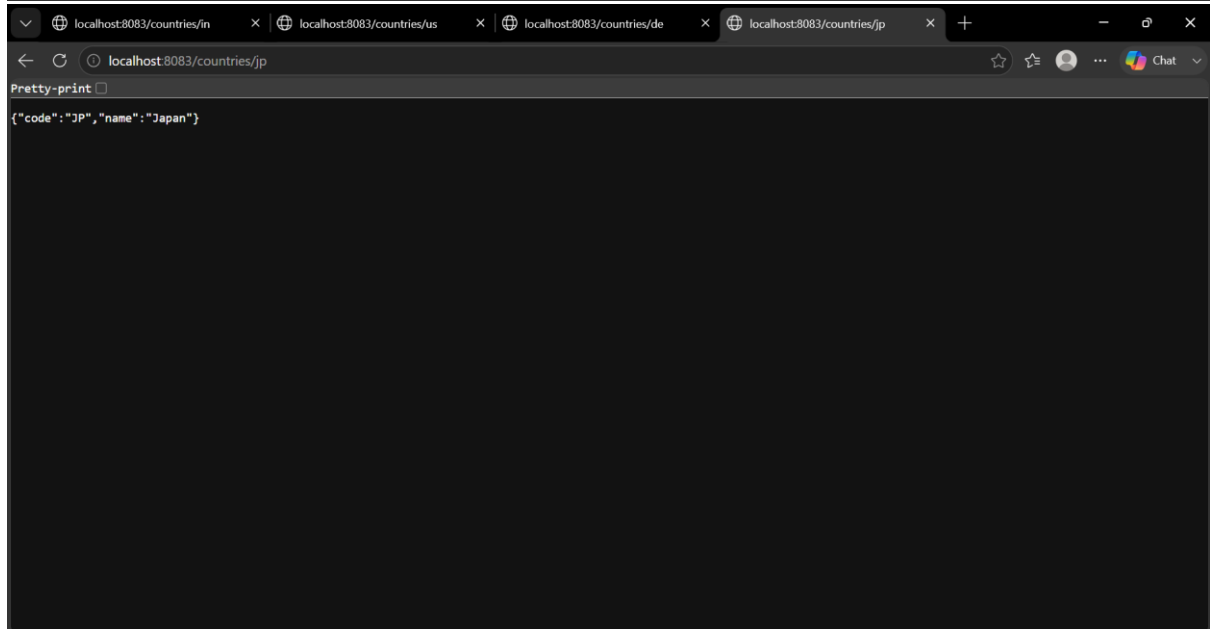
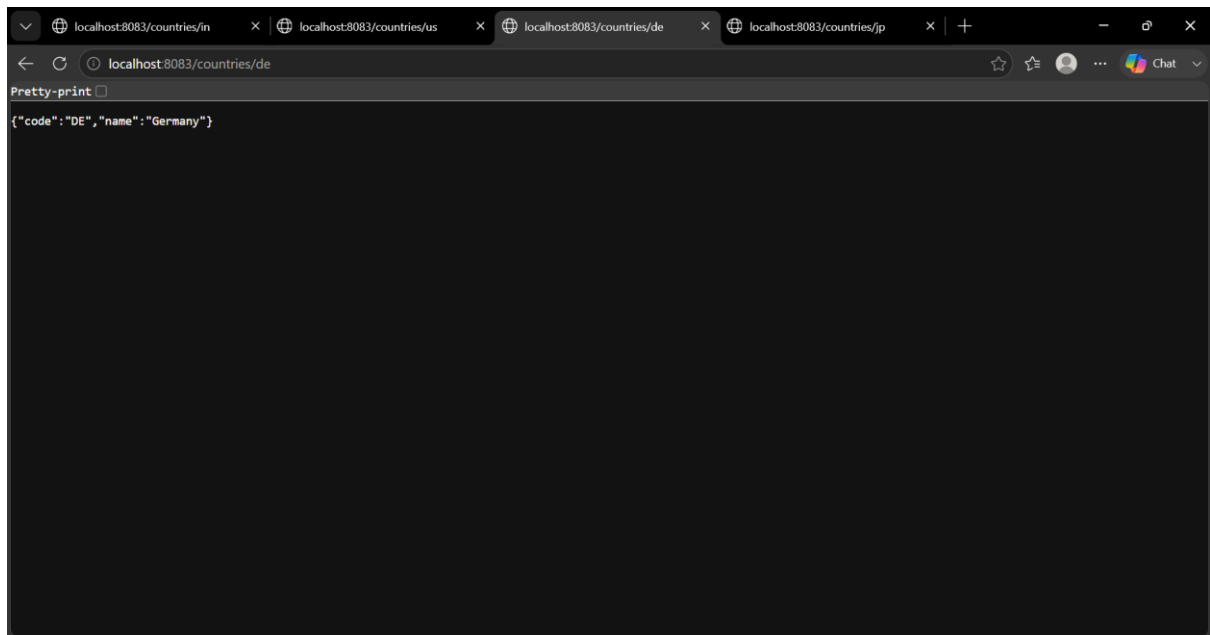
*****
APPLICATION FAILED TO START
*****

Description:
Web server failed to start. Port 8083 was already in use.

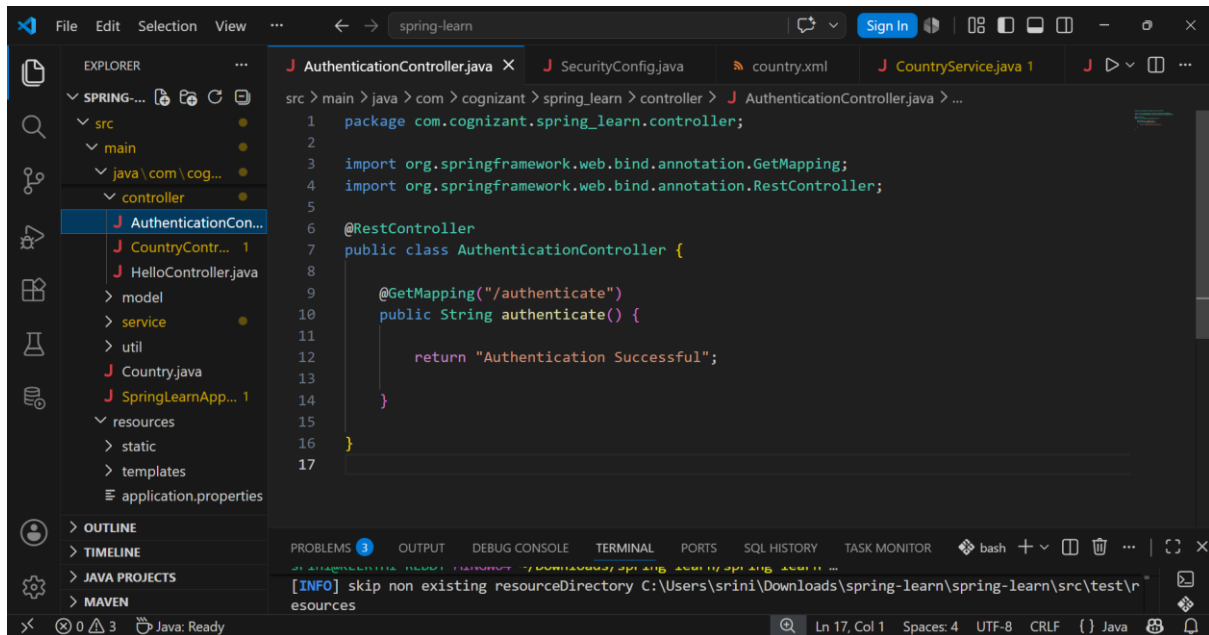
Action:
Identify and stop the process that's listening on port 8083 or configure this application to listen on another port.

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.118 s
[INFO] Finished at: 2026-07-01T14:41:46+05:30
[INFO] -----
spring@KEERTHI-REDDY HINGW64: ~/Downloads/spring_learn/spring_learn
$
```





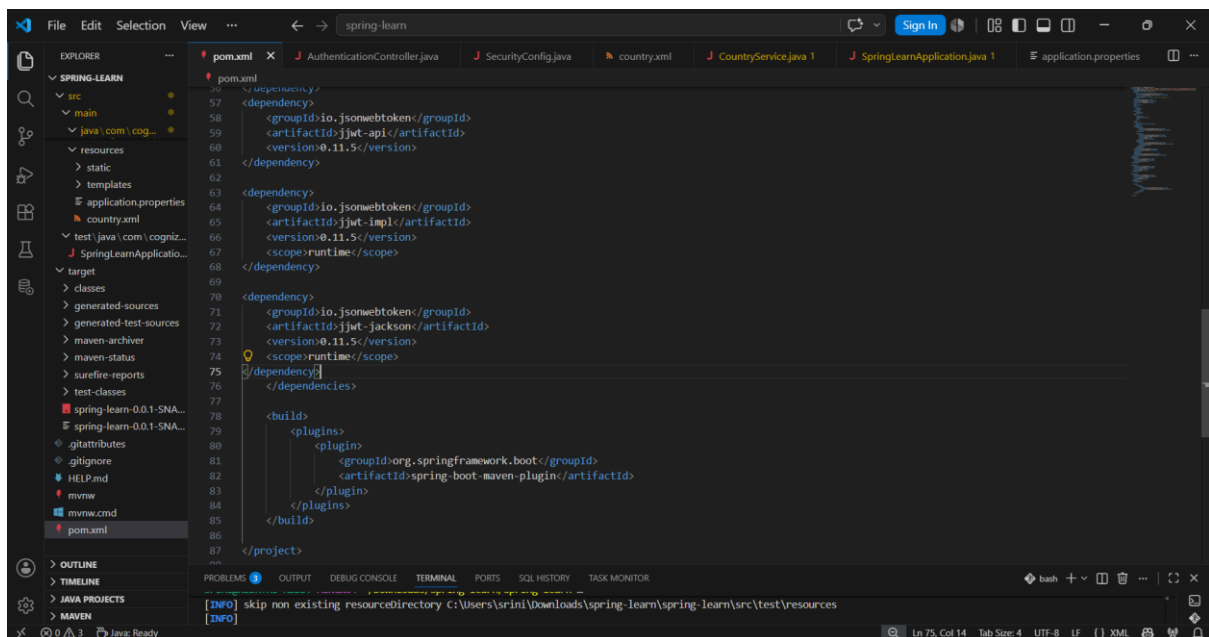
Create authentication service that returns JWT



This screenshot shows an IDE window with the file `AuthenticationController.java` open. The Explorer on the left shows the project structure with a `controller` package. The code in the editor is as follows:

```
src > main > java > com > cognizant > spring_learn > controller > J AuthenticationController.java > ...
1  package com.cognizant.spring_learn.controller;
2
3  import org.springframework.web.bind.annotation.GetMapping;
4  import org.springframework.web.bind.annotation.RestController;
5
6  @RestController
7  public class AuthenticationController {
8
9      @GetMapping("/authenticate")
10     public String authenticate() {
11
12         return "Authentication Successful";
13     }
14 }
15
16
17
```

The bottom of the IDE shows the PROBLEMS panel with an [INFO] message: `skip non existing resourceDirectory C:\Users\sri\Downloads\spring-learn\spring-learn\src\test\resources`.



This screenshot shows the same IDE window with the `pom.xml` file open. The Explorer on the left shows the project structure with a `resources` directory. The code in the editor is as follows:

```
pom.xml
57 <dependency>
58 <groupId>io.jsonwebtoken</groupId>
59 <artifactId>jjwt-api</artifactId>
60 <version>0.11.5</version>
61 </dependency>
62
63 <dependency>
64 <groupId>io.jsonwebtoken</groupId>
65 <artifactId>jjwt-impl</artifactId>
66 <version>0.11.5</version>
67 <scope>runtime</scope>
68 </dependency>
69
70 <dependency>
71 <groupId>io.jsonwebtoken</groupId>
72 <artifactId>jjwt-jackson</artifactId>
73 <version>0.11.5</version>
74 <scope>runtime</scope>
75 </dependency>
76
77 <build>
78 <plugins>
79 <plugin>
80 <groupId>org.springframework.boot</groupId>
81 <artifactId>spring-boot-maven-plugin</artifactId>
82 </plugin>
83 </plugins>
84 </build>
85
86 </project>
87
```

The bottom of the IDE shows the PROBLEMS panel with two [INFO] messages: `skip non existing resourceDirectory C:\Users\sri\Downloads\spring-learn\spring-learn\src\test\resources` and `skip non existing resourceDirectory C:\Users\sri\Downloads\spring-learn\spring-learn\src\test\resources`.

